Ayomide Adetiloye

Location: Birmingham UK | **Mobile:** +44 (0)7486058123 | **Email:** Ayomide.Adetiloye@outlook.com

MEng Mechatronics and Robotics student with hands on experience in turning designs into working hardware and machinery integrating sensors, actuators and embedded control into systems that perform reliably. Strong analytical and problem solving skills with keen interest and good understanding in how control principles are applied across different systems. Eager to build a career in engineering and technology, applying my skills to real world problems and learning from experienced engineers

Education

University of Birmingham – MEng Mechatronics and Robotics Engineering: (Sep 24 – Present)

Modules: Mechatronics & Control Engineering, Digital Electronics and Electrical Machines, Electronic Circuits Devices and Electromagnetics, Introduction to Computing for Engineers, Engineering Mathematics

Gravesend Grammar School (Oct 20 – Aug 24)
A-Levels: Mathematics (A), Physics (B) and Chemistry (B)
GCSEs: 2A*s, 6As, 1B

Technical Skills

- **Programming:** C/C++, Python, MATLAB/Simulink, Arduino, HTML, CSS, JavaScript
- **Embedded Systems & Control:** microcontroller programming, sensor interfacing and signal conditioning, motor and actuator control, real time data acquisition.
- **Design & Simulation:** CAD (Fusion 360), PCB design (EasyEDA, Altium)
- **Manufacturing:** 3D printing, laser cutting, lathe and milling machine operation, soldering and rework
- **Analysis:** data analysis & visualization, engineering mathematics, statistics

Experience

Internship – Petrofac (Aug 23)

- Worked alongside experienced engineers in an onshore support role for offshore operations, gaining firsthand exposure to professional engineering practice in a safety critical industry
- Contributed technical input to project meetings, engaging with progress tracking, scheduling and maintenance planning
- Managed maintenance documentation and scheduling for offshore rig systems, building skills in reliability and systems engineering

Virtual Work Experience – Johnson & Johnson MedTech | Robotics and Control (Sep 26)

- Diagnosed a simulated response time fault in a surgical robotic arm's control systems by analysing Python control code, tracing the delay back to specific software inefficiencies
- Designed and validated targeted hardware/software modifications (a functional actuator coupling and a redundant sensor fusion fix) utilising step based testing, improving simulated response times by 15% and durability score by 20%
- Produced structured technical design proposals with annotated diagrams, metric data and recommendations for long term reliability

Secretary, Engineering Society (Jan 24 – Aug 24)

- Acted as the central point of contact for society members ensuring quick communication of updates, events and opportunities.
- Maintained detailed and accurate records and correspondence, supporting smooth day to day running of the society and continuity between committees.

Volunteer Tutor – Whitehill Primary School (Sep 23 – Dec 23)

- Broke down complex problems into manageable steps adapting my teaching approach based on individual student needs.
- Planned and delivered engaging activities whilst effectively managing time to ensure students were focused and on track.

Combined Cadet Force (Aug 18 – Oct 20)

- Developed teamwork, discipline and confidence through structured drills and group exercises
- Stayed composed under pressure, adapting plans and solving problems when situations go astray
- Learned map reading and navigation techniques to plan routes and traverse unfamiliar terrain

Projects & Achievements

- **Robotic Arm –** Designing and developing a 3D printed a multi jointed robotic Arm with an extensive range of motion. Engineering custom cycloidal actuators to drive joint movement with high torque and low backlash
- **Arduino Security System –** Built a fully functional security system integrating sensors buzzers and keypad authentication.
- **Weight Sensing Glove –** Developing a wearable glove using force sensitive resistors to measure and output real time grip force exerted by the wearer.
- **De-icing agent dispenser –** engineered an actuator and relay controlled dispensing mechanism, designed for potential deployment on road surfaces.
- **Assistant Head of Key Stage 4 Award –** An award given to the top performing student across the entire year, recognized for academic excellence, leadership and consistent diligence.